

A24 OnchipDS

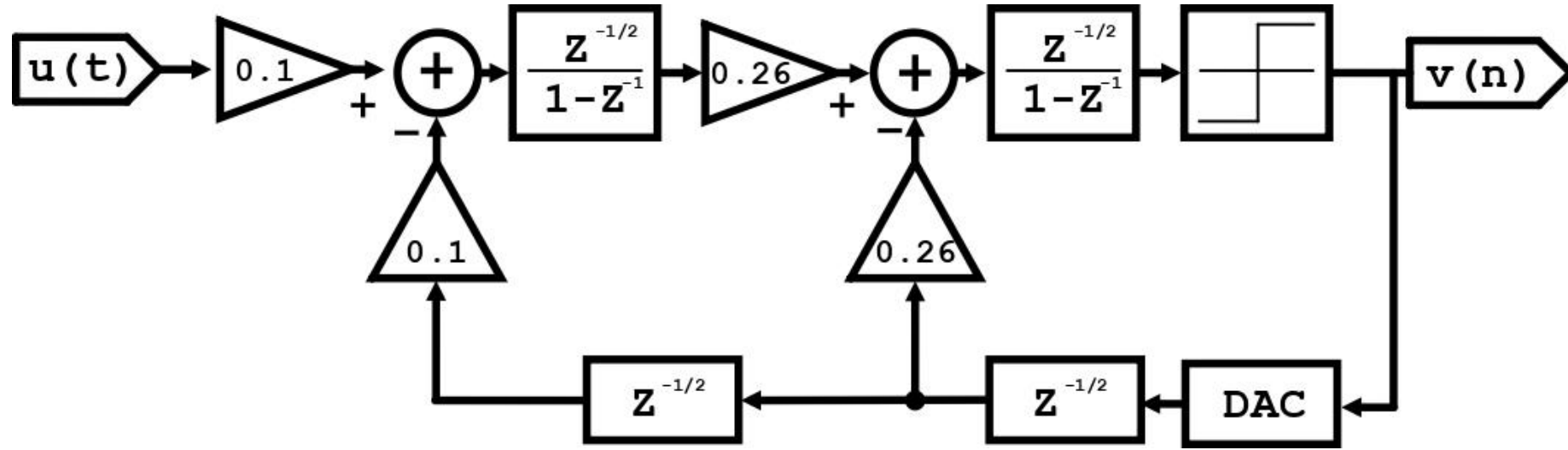
Title: A Ring-Amplifier Based Delta-Sigma Modulator

Discord name	Affiliation	Experience	Role
AlexM	Universidad Industrial de Santander (UIS)	MSc Student	Team lead
RicardoV		MSc Student	Member

*And some occasional contributors

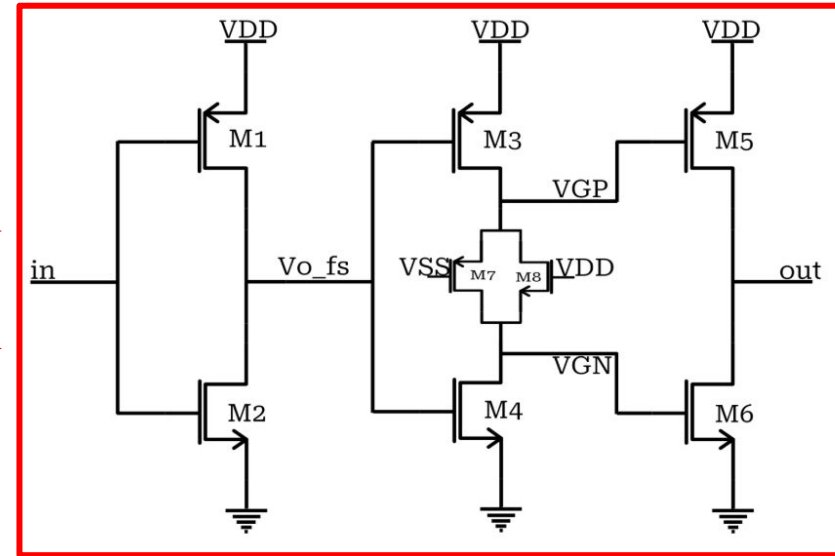
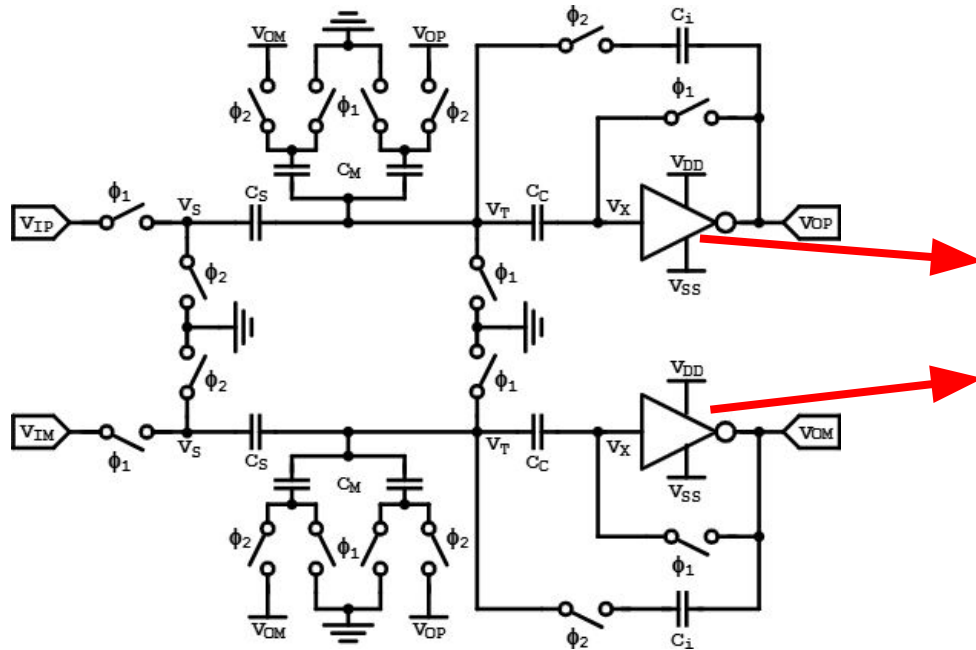
Project Goal:

- To Design a Low-Power High-Resolution Ring-Amplifier Based 2nd-Order Delta-Sigma ($\Delta\Sigma$) Modulator for audio applications, with an 20 kHz bandwidth.



Design Highlight:

- SC Loop filter implemented with Ring-Amplifier allows for **extreme supply voltage reduction, wide output swing** and **faster response**.



References:

- [1] Y. Chae and G. Han, "Low Voltage, Low Power, Inverter-Based Switched-Capacitor Delta-Sigma Modulator," in IEEE Journal of Solid-State Circuits, vol. 44, no. 2, pp. 458-472, Feb. 2009, doi: 10.1109/JSSC.2008.2010973
- [2] J. Lagos, B. P. Hershberg, E. Martens, P. Wambacq and J. Craninckx, "A 1-GS/s, 12-b, Single-Channel Pipelined ADC With Dead-Zone-Degenerated Ring Amplifiers," in IEEE Journal of Solid-State Circuits, vol. 54, no. 3, pp. 646-658, March 2019, doi: 10.1109/JSSC.2018.2889680

Questions:

- Is there a **speed limit** for I/O digital PADs? (expected)
- Could each project have **an independent power supply**, separating the ring with circuit breakers?

